Multiobjective Programming and Planning

Fundamentals of Multiobjective Optimization

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Today's Topics

1. **Motivation for Multiobjective Optimization**
2. **Key Concepts and Definitions**
3. **Multiobjective Knapsack Problem (MOKP) Example**
4. **Challenges in Multiobjective Optimization**

References

- Cohon, J.L. (1978) - Multiobjective Programming and Planning
- Czyzak and Jaszkiewicz (1998)
- Deb, K. (2001) - Multi-Objective Optimization Using Evolutionary Algorithms

Motivation for Multiobjective Approaches

According to Czyzak and Jaszkiewicz (1998)

- Many decision problems involve **multiple stakeholders**
- Each stakeholder has **different criteria and preferences**
- Single-objective approaches may lead to **inadequate solutions**

According to Cohon (1978)

- Multiobjective programming = **generalization** of traditional single-objective approaches

Key Advantages Identified by Cohon (1978)

1. **More appropriate decision rules** for stakeholders involved in the planning and decision-making process
2. **Broader range** of intervention alternatives is usually identified
3. **More realistic** mathematical models and analysts' perceptions when multiple objectives are incorporated

Problem Formulation

Traditional Single-Objective

$$\text{Maximize: } f(x)$$
$$\text{Subject to: } x \in D$$

Multiobjective

$$\text{Maximize: } [f_1(x), f_2(x), \ldots, f_k(x)]$$
$$\text{Subject to: } x \in D$$

Note

The multiobjective formulation seeks to optimize **multiple conflicting objectives** simultaneously.

Transforming Multiobjective into Single-Objective

Weighted Combination

- Vector of weights: $w = [w_1, w_2, \ldots, w_k]$
- Transformation: Maximize $\sum_{j=1}^{k} w_j f_j(x)$

Problems with Fixed Weights

- Determining appropriate weights is **difficult and time-consuming**
- Fixed weights may **fail to accurately represent** stakeholder preferences
- Weight sensitivity can affect solution quality

Definition of Non-dominated Solutions

Definition (Deb, 2001)

A solution $x \in D$ is said to be **non-dominated (Pareto-optimal)** if there is **no** solution $x' \in D$ such that:

$$f_j(x') \geq f_j(x), \quad \forall j$$

and

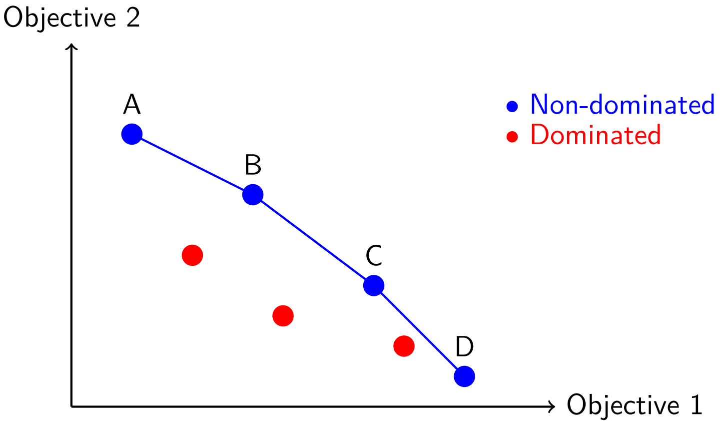
$$f_j(x') > f_j(x)$$

for at least one objective j.

Graphical Representation

Non-dominated solutions form the **Pareto frontier** — a curve or surface in objective space.

Graphical Representation for Maximization



Key Insight

Improving one objective typically worsens another — this is the **trade-off** relationship.

Classical Knapsack Problem

Problem Description

- Set of n items
- Each item j: profit c_j and weight w_j
- Knapsack capacity: W
- Goal: Select subset of items to **maximize total profit**

Decision Variables

$$x_j = \begin{cases} 1, & \text{if item } j \text{ is included in the knapsack} \\ 0, & \text{otherwise} \end{cases}$$

Integer Linear Programming Model

Mathematical Model

$$\text{Maximize:} \quad \sum_{j=1}^{n} c_j x_j \tag{1}$$

$$\text{Subject to:} \quad \sum_{j=1}^{n} w_j x_j \leq W \tag{2}$$

$$x_j \in \{0, 1\}, \quad j = 1, \ldots, n \tag{3}$$

- Objective: Maximize total profit of selected items
- Constraint: Total weight cannot exceed knapsack capacity

Two-Objective Formulation

Objectives

$$\text{Maximize: } z_1(x) = \sum_{j=1}^{n} c_j x_j \tag{4}$$

$$\text{Maximize: } z_2(x) = W - \sum_{j=1}^{n} w_j x_j \tag{5}$$

Constraints

$$x_j \in \{0, 1\}, \quad j = 1, \ldots, n$$

Interpretation

- z_1: Maximize total profit
- z_2: Maximize remaining knapsack capacity

Problem Instance

Knapsack Capacity: $W = 100$

Item	1	2	3	4	5
w_j	25	15	14	25	15
c_j	80	120	80	15	30

Table: Data for the MOKP example

Combinatorial Complexity

Total number of possible solutions: $2^n = 2^5 = 32$

Why is MOKP Challenging?

Computational Challenges

- **Combinatorial optimization** problem (NP-hard)
- Multiple objectives add significant complexity
- Exact methods are often impractical for large instances

Metaheuristics Approach

- Balance **diversification** and **intensification**
- Generate **diverse set** of non-dominated solutions
- Ensure solutions are **close to true Pareto frontier**

Goals in Multiobjective Metaheuristic Design

1. Diversity Preservation

- Spread solutions across the Pareto frontier
- Cover different trade-off regions
- Avoid clustering in one area

2. Convergence

- Solutions close to true Pareto frontier
- Avoid dominated solutions
- Maximize solution quality

Dual Objective

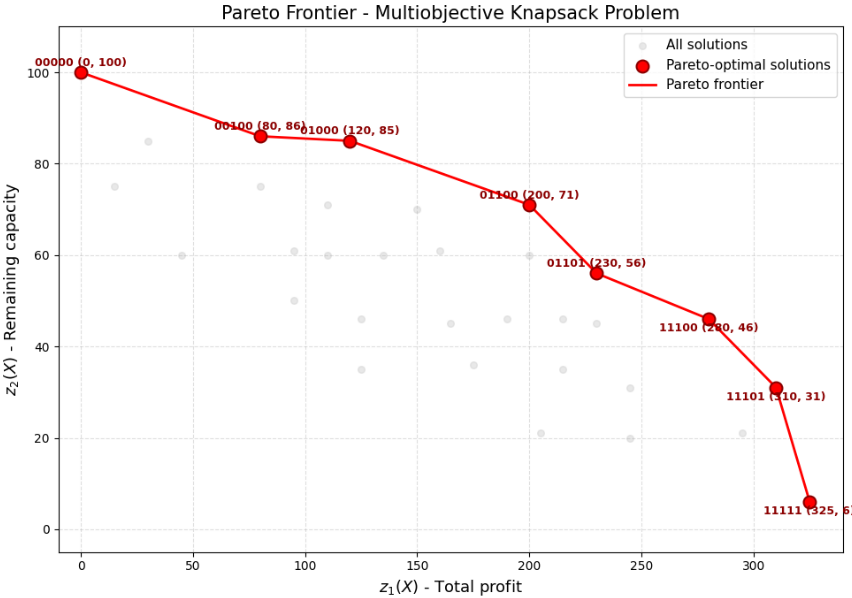
Both diversity and convergence are **fundamental** for effective multiobjective optimization.

Complete Solution Table (Non-dominated in Bold)

Sol.	z_1	z_2	Sol.	z_1	z_2	Sol.	z_1	z_2	Sol.	z_1	z_2
1	**0**	**100**	9	95	50	**17**	**280**	**46**	25	230	56
2	80	75	10	110	60	18	215	35	26	125	46
3	**120**	**85**	**11**	**200**	**71**	19	230	45	27	295	21
4	**80**	**86**	12	135	60	20	175	36	28	245	31
5	15	75	13	150	70	21	190	46	29	205	21
6	30	85	14	95	61	22	125	35	30	245	20
7	200	60	15	110	71	23	215	46	**31**	**310**	**31**
8	160	61	16	45	60	24	165	45	**32**	**325**	**6**

Table: All 32 possible solutions for the MOKP example. Non-dominated solutions are highlighted in bold.

Graphical Representation of Non-dominated Solutions



Comparison Examples

Solution 2 vs. Solution 4

- Solution 2: $z_1 = 80$, $z_2 = 75$
- Solution 4: $z_1 = 80$, $z_2 = 86$
- Solution 4 dominates Solution 2 (higher z_2, equal z_1)

Solution 7 vs. Solution 11

- Solution 7: $z_1 = 200$, $z_2 = 60$
- Solution 11: $z_1 = 200$, $z_2 = 71$
- Solution 11 dominates Solution 7 (higher z_2, equal z_1)

Solution 12 vs. Solution 11

- Solution 12: $z_1 = 135$, $z_2 = 60$
- Solution 11: $z_1 = 200$, $z_2 = 71$
- Solution 11 dominates Solution 12 (higher z_1 and higher z_2)

Key Observations

Solution Space

- Total possible solutions: 32
- Only **7 solutions** are non-dominated (22%)
- Non-dominated solutions represent **Pareto frontier**

Dominance Property

- Any dominated solution can be **improved**
- Improvement in at least one objective
- Without worsening the other objective(s)

Academic Perspective

According to Ulungu et al. (1999)

- Multiobjective combinatorial optimization problems are **particularly difficult** to solve using exact methods
- NP-hard nature of many combinatorial problems

Current Research Directions

- **Highly active research area**
- **Metaheuristics** are widely used
- Important contributions: Viana (2004), Dias (2005)

Comparative Analysis

Single-Objective

- Balance **diversification** and **intensification**
- Locate high-quality solutions
- Within acceptable computational times

Multiobjective

- Generate **diverse set** of non-dominated solutions
- Ensure solutions are **close** to true Pareto frontier
- **Both diversity and convergence** are fundamental

Summary of Main Concepts

1. Multiobjective optimization handles **multiple conflicting criteria**
2. **Pareto frontier** represents optimal trade-offs between objectives
3. **MOKP** illustrates multiobjective combinatorial optimization
4. **Metaheuristics** are essential for solving large problems
5. **Diversity** and **convergence** are dual goals in multiobjective optimization

Main Reference

Cohon, J.L. (1978). *Multiobjective Programming and Planning*. Academic Press.

A-priori Methods

Key Concept

Preferences are articulated **before** solution generation

Methods to be Covered

- Lexicographic Method
- Goal Programming
- Utility Function Methods
- Weighted Sum Method

Reading

Based on Cohon (1978) - Chapter 7

References

- Cohon, J.L. (1978) - *Multiobjective Programming and Planning*. New York: Academic Press, 1978.
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- Deb, K. (2001) - *Multi-Objective Optimization Using Evolutionary Algorithms*
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- Ulungu, E. L.; Teghem, J. F. P. H.; Fortemps, P. H.; Tuyttens, D. (1999) - MOSA method: a tool for solving multiobjective combinatorial optimization problems. *Journal of Multi-Criteria Decision Analysis*, 8(4), 221.
- Viana, A. (2004) - Metaheuristics for the unit commitment problem: the constraint oriented neighbourhoods search strategy. Ph.D. Thesis (Doctorate in Electrical Engineering and Computers). Faculty of Engineering, University of Porto, Porto.